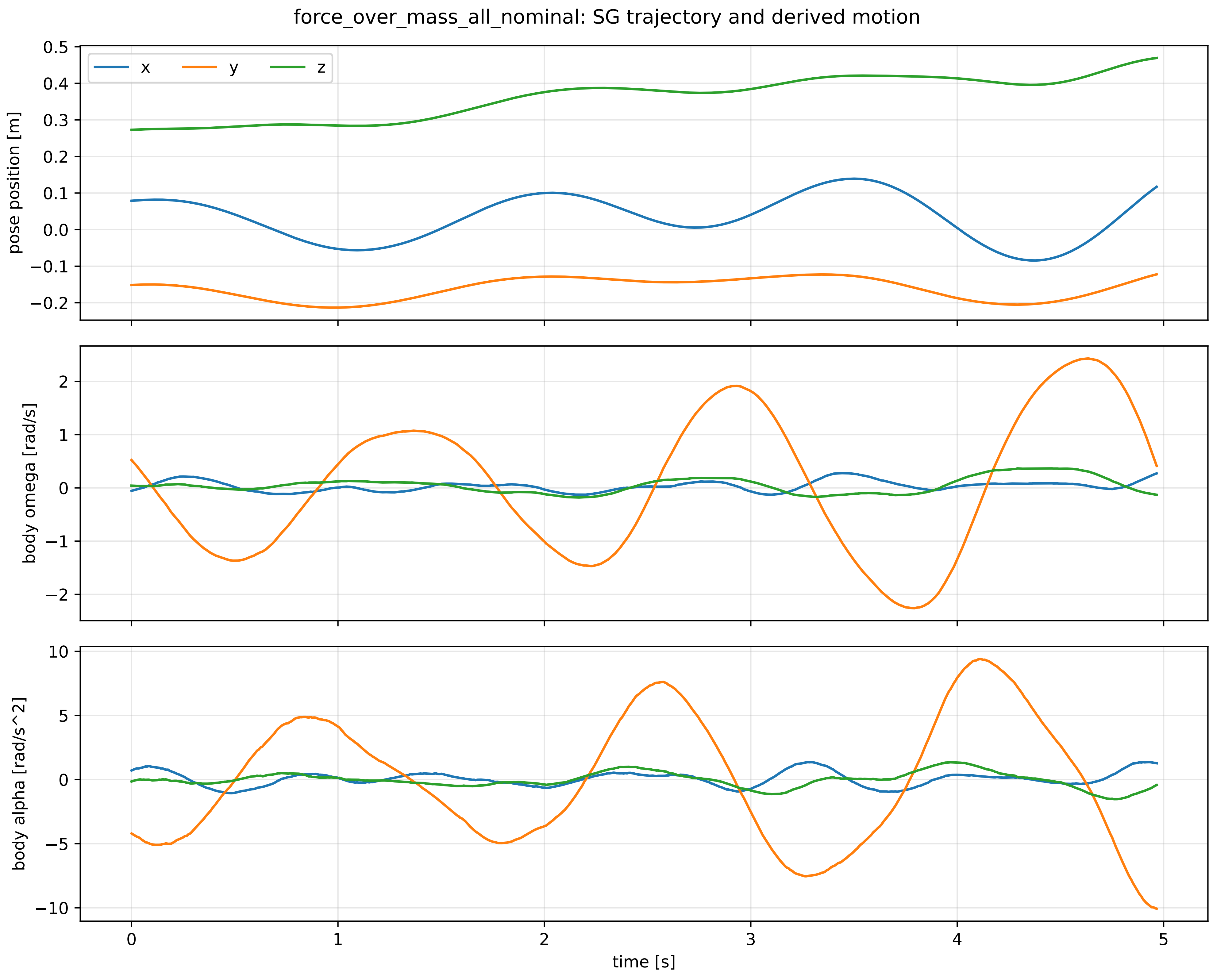
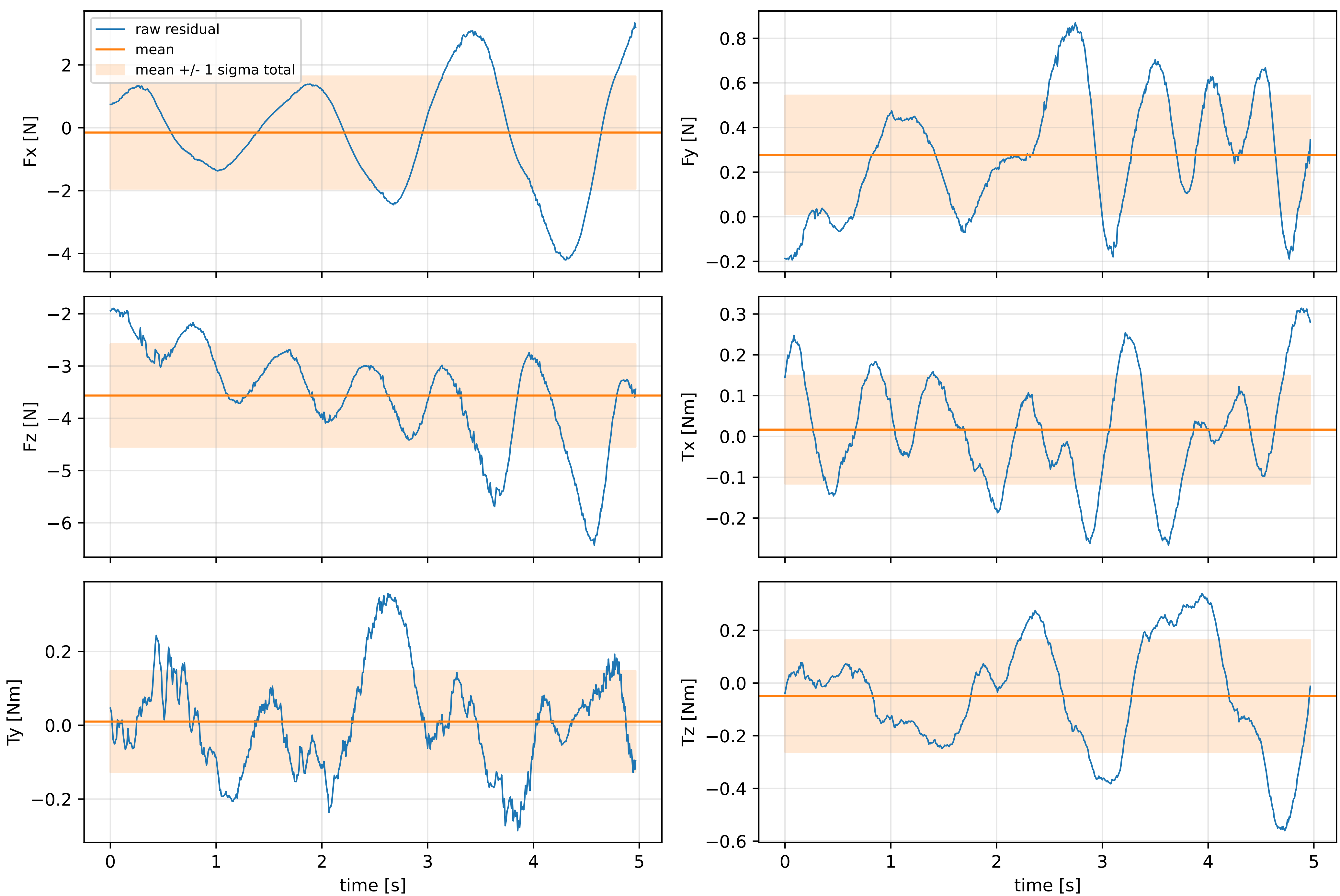
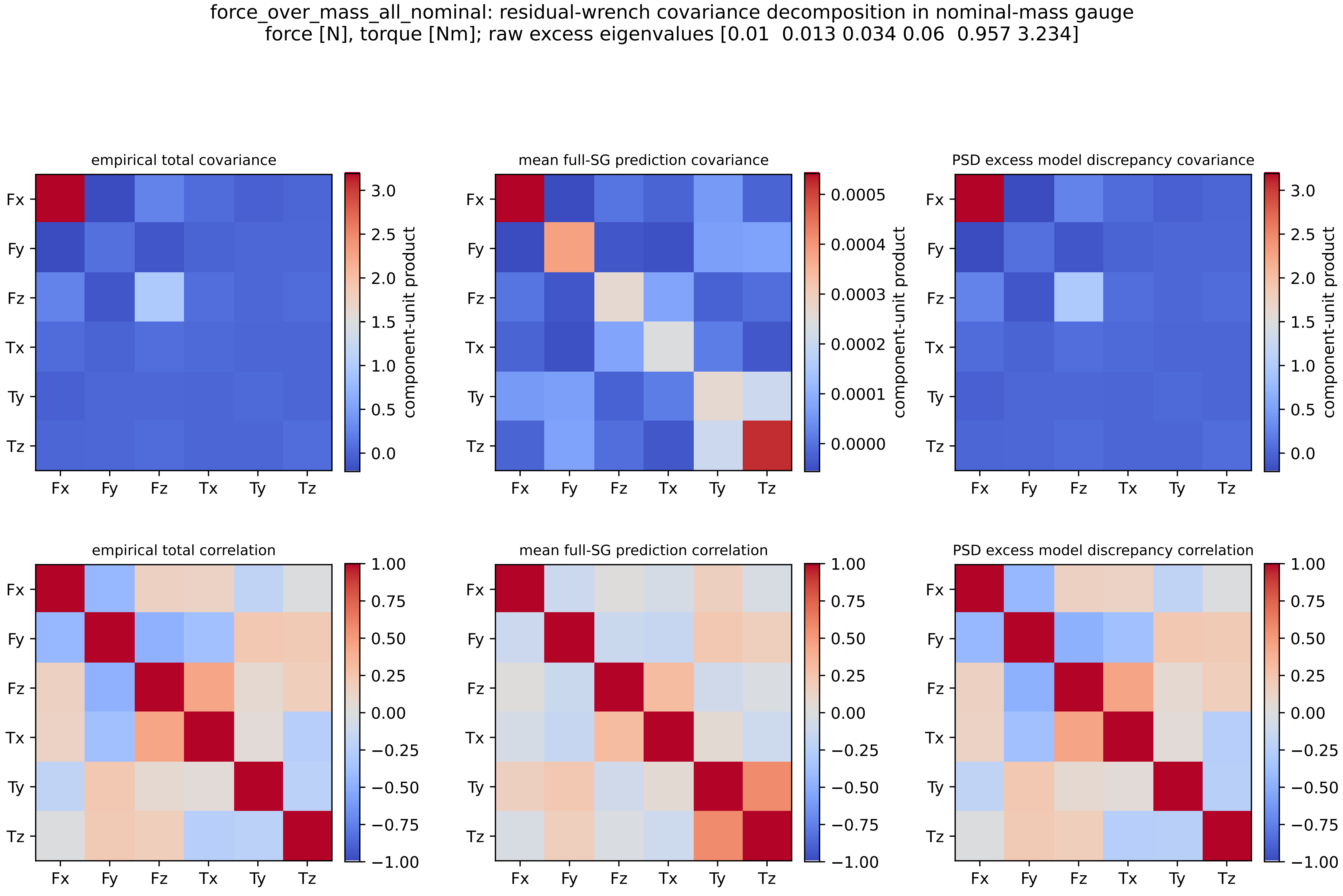


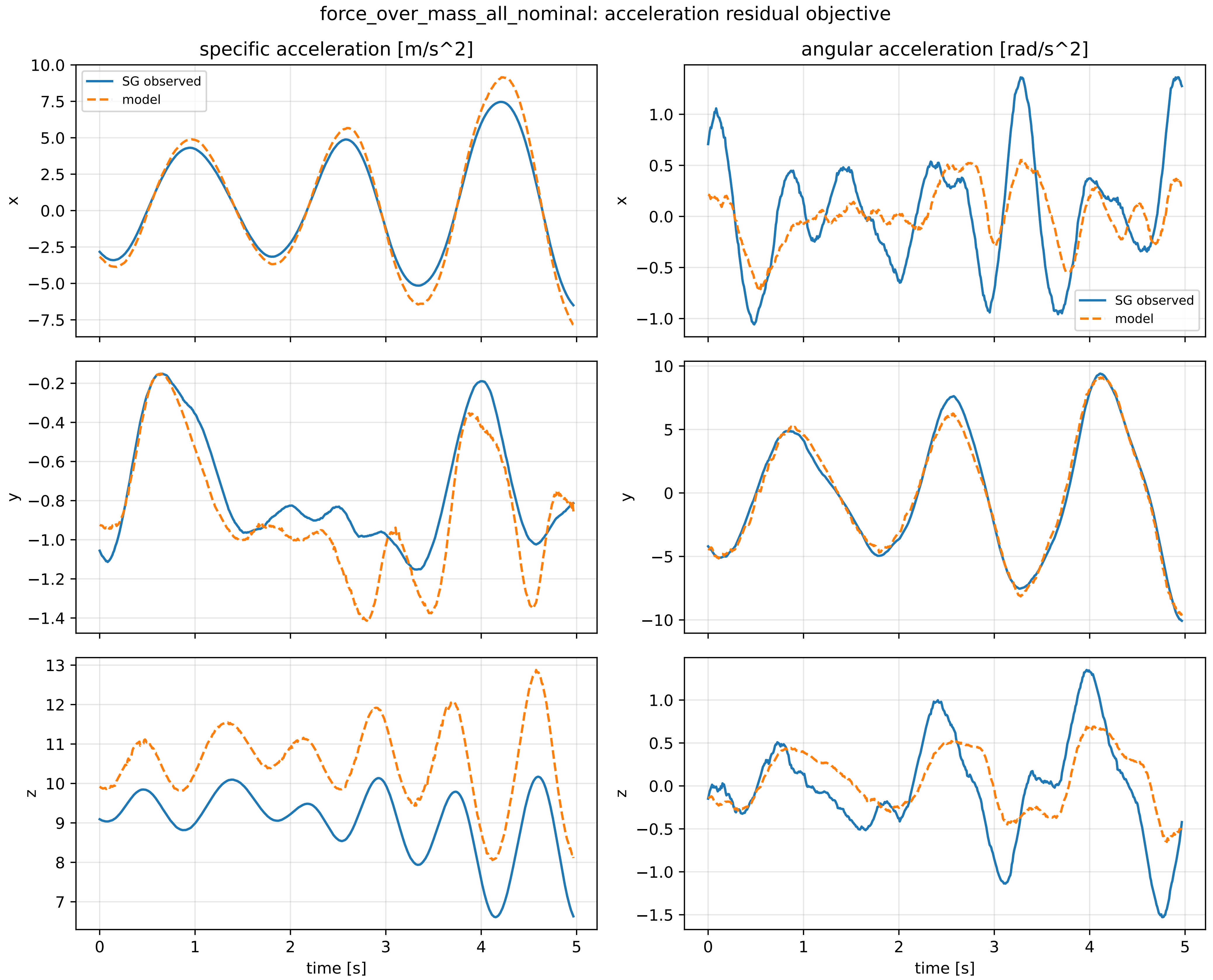
force_over_mass_all_nominal: SG trajectory and derived motion
x
y
z
pose position [m]
body omega [rad/s]
body alpha [rad/s^2]
time [s]

force_over_mass_all_nominal: raw Newton--Euler residual wrench (not trajectory-fitted external wrench)
mass gauge fixed to nominal mass = 2.35159759 kg

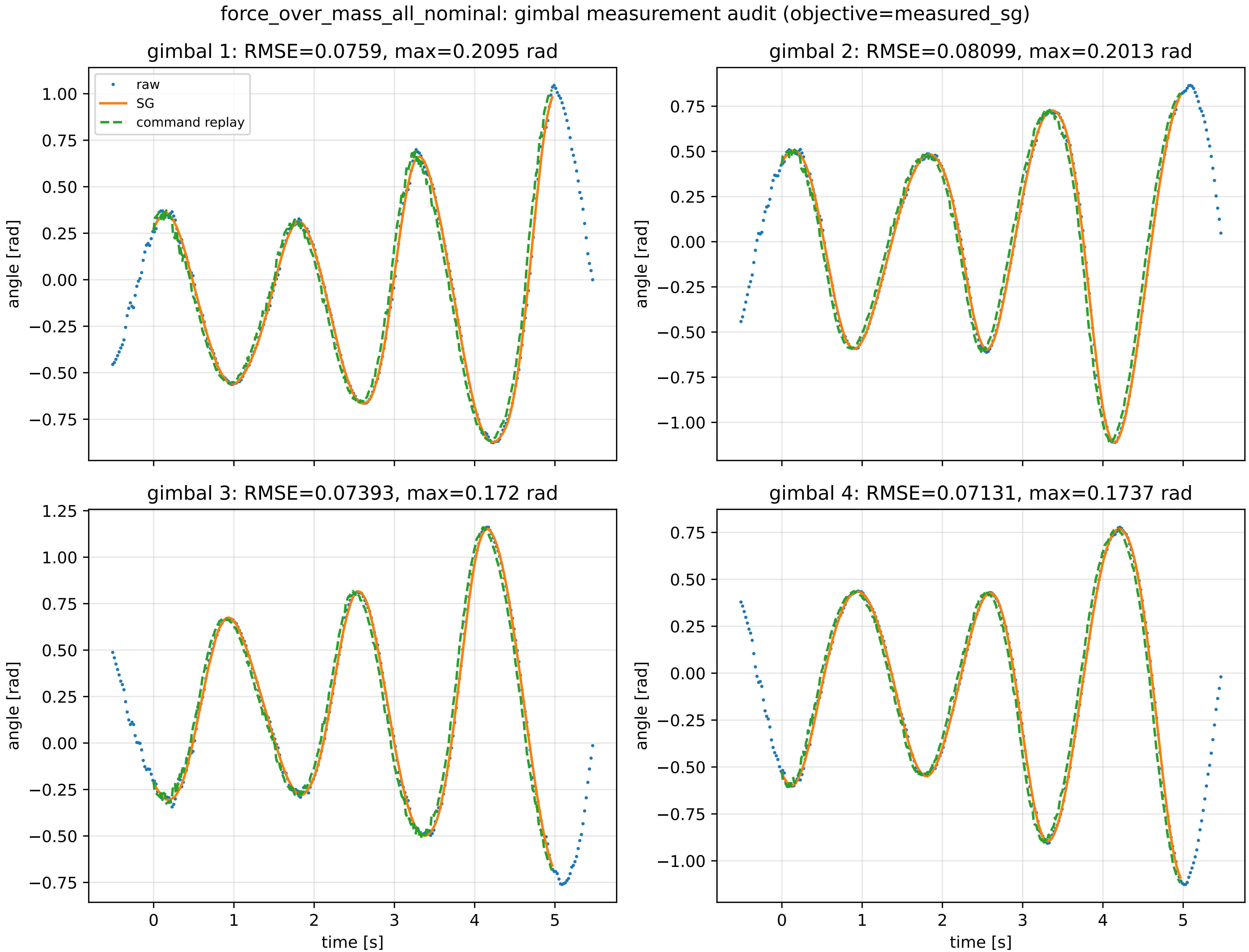


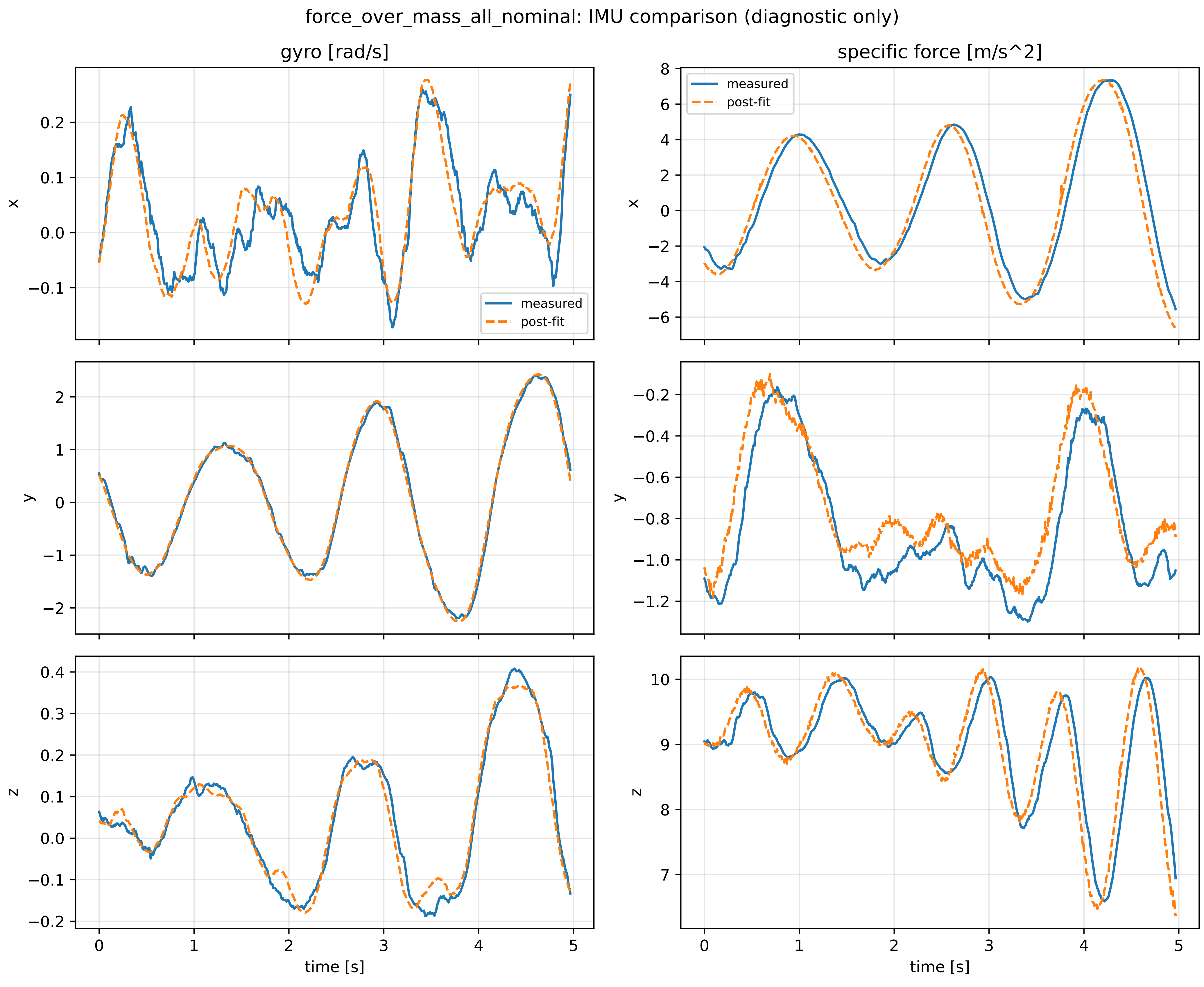
vector RMS about zero: force 4.1263 N, torque 0.28987 Nm; component std about mean: force [1.7868 0.2659 0.9826], torque [0.1326 0.1372 0.2119]

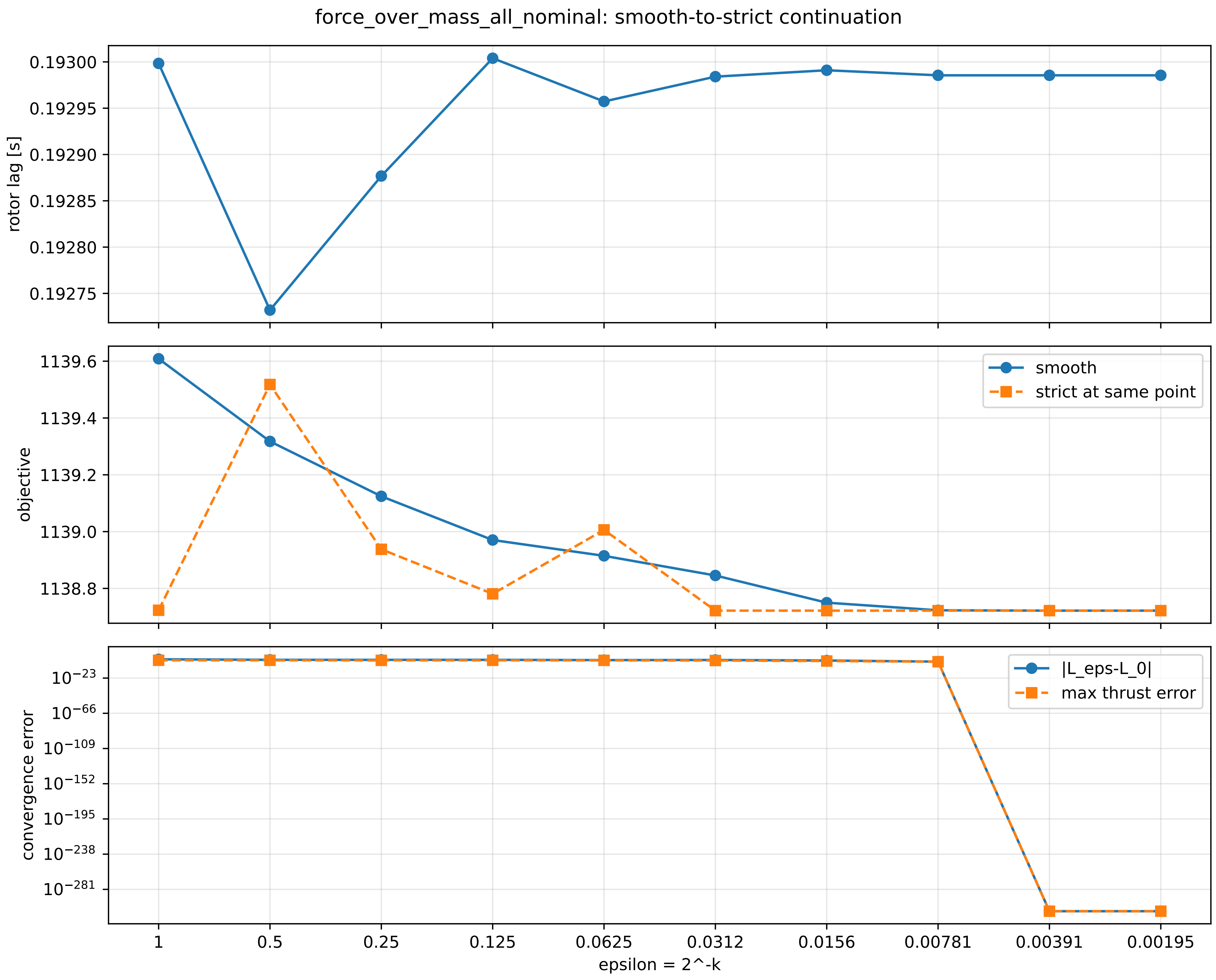

force_over_mass_all_nominal: residual-wrench covariance decomposition in nominal-mass gauge
force [N], torque [Nm]; raw excess eigenvalues [0.01 0.013 0.034 0.06 0.957 3.234]
empirical total covariance
mean full-SG prediction covariance
PSD excess model discrepancy covariance
empirical total correlation
mean full-SG prediction correlation
PSD excess model discrepancy correlation
component-unit product
Fx Fy Fz Tx Ty Tz



force_over_mass_all_nominal: acceleration residual objective
specific acceleration [m/s^2]
angular acceleration [rad/s^2]
SG observed
model
x
y
z
time [s]

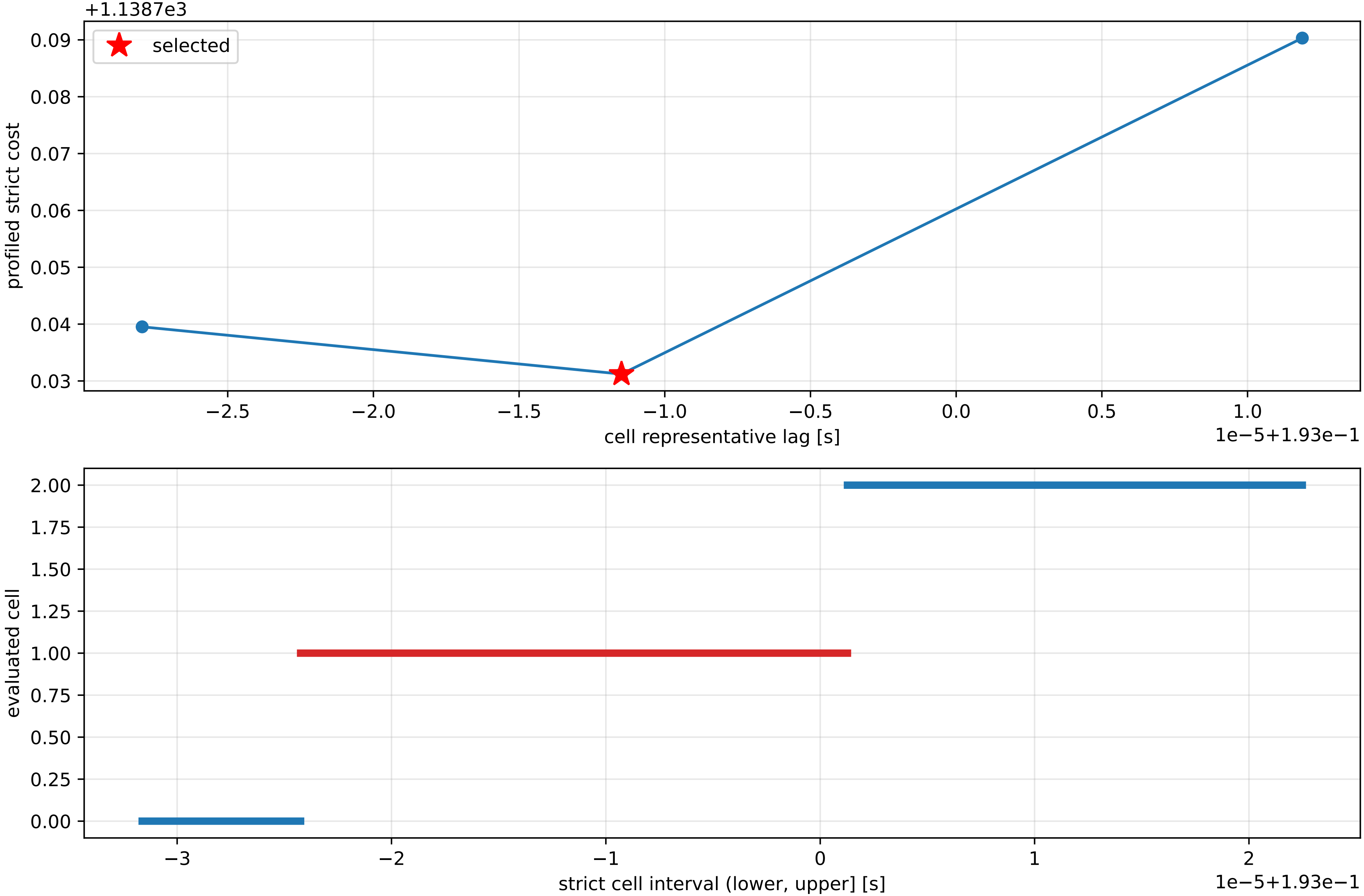

force_over_mass_all_nominal: gimbal measurement audit (objective=measured_sg)
gimbal 1: RMSE=0.0759, max=0.2095 rad
gimbal 2: RMSE=0.08099, max=0.2013 rad
gimbal 3: RMSE=0.07393, max=0.172 rad
gimbal 4: RMSE=0.07131, max=0.1737 rad
raw
SG
command replay
angle [rad]
time [s]


force_over_mass_all_nominal: IMU comparison (diagnostic only)
gyro [rad/s]
specific force [m/s^2]
measured
post-fit
x
y
z
time [s]



force_over_mass_all_nominal: smooth-to-strict continuation
rotor lag [s]
0.19300
0.19295
0.19290
0.19285
0.19280
0.19275
objective
1139.6
1139.4
1139.2
1139.0
1138.8
smooth
strict at same point
convergence error
10^{-23}
10^{-66}
10^{-109}
10^{-152}
10^{-195}
10^{-238}
10^{-281}
|L_eps-L_0|
max thrust error
1
0.5
0.25
0.125
0.0625
0.0312
0.0156
0.00781
0.00391
0.00195
epsilon = 2^-k

force_over_mass_all_nominal: exact strict-ZOH lag cells

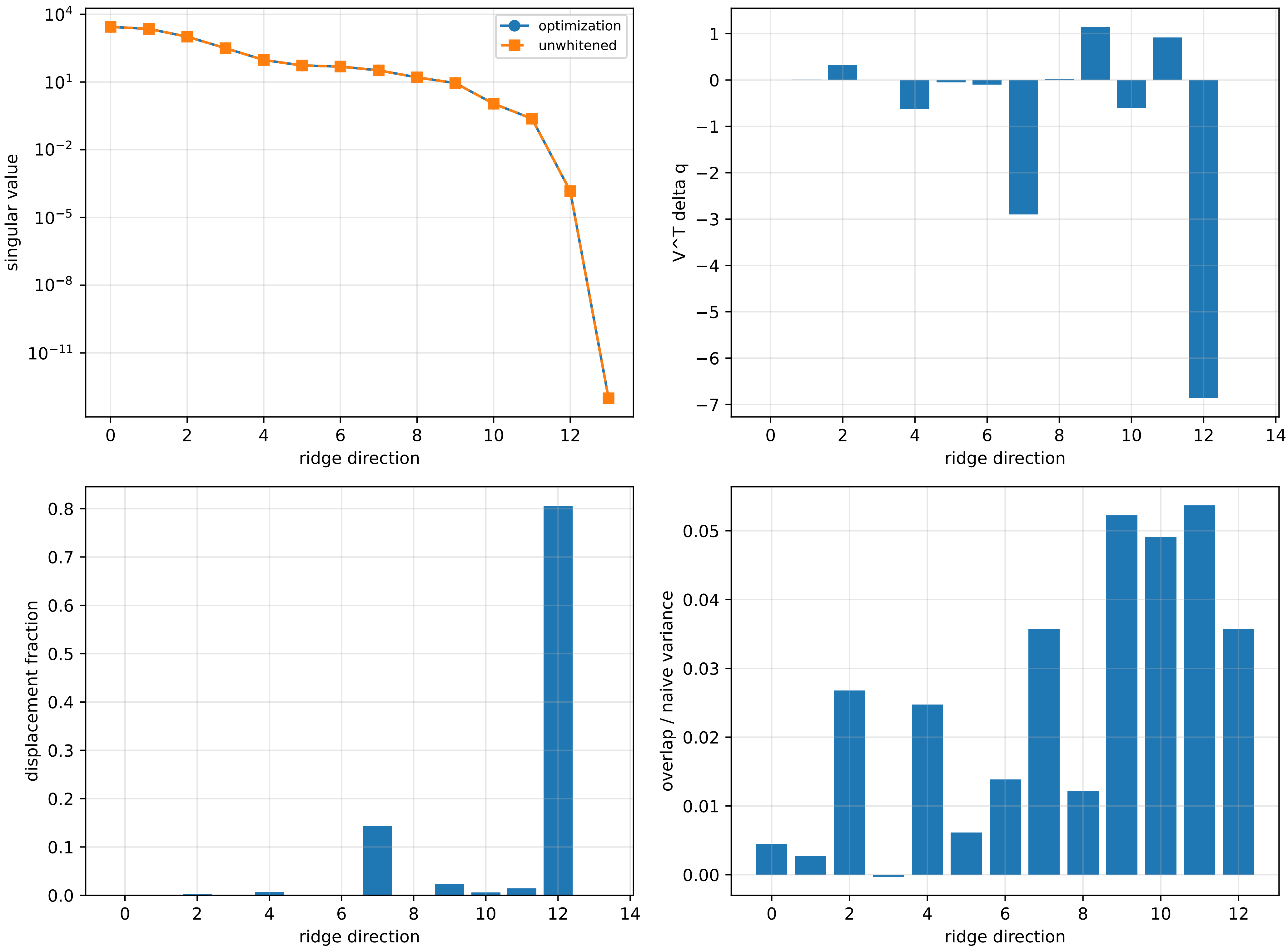


selected (0.19297576, 0.19300127] s; final smooth 0.192985528 s; data boundary=False

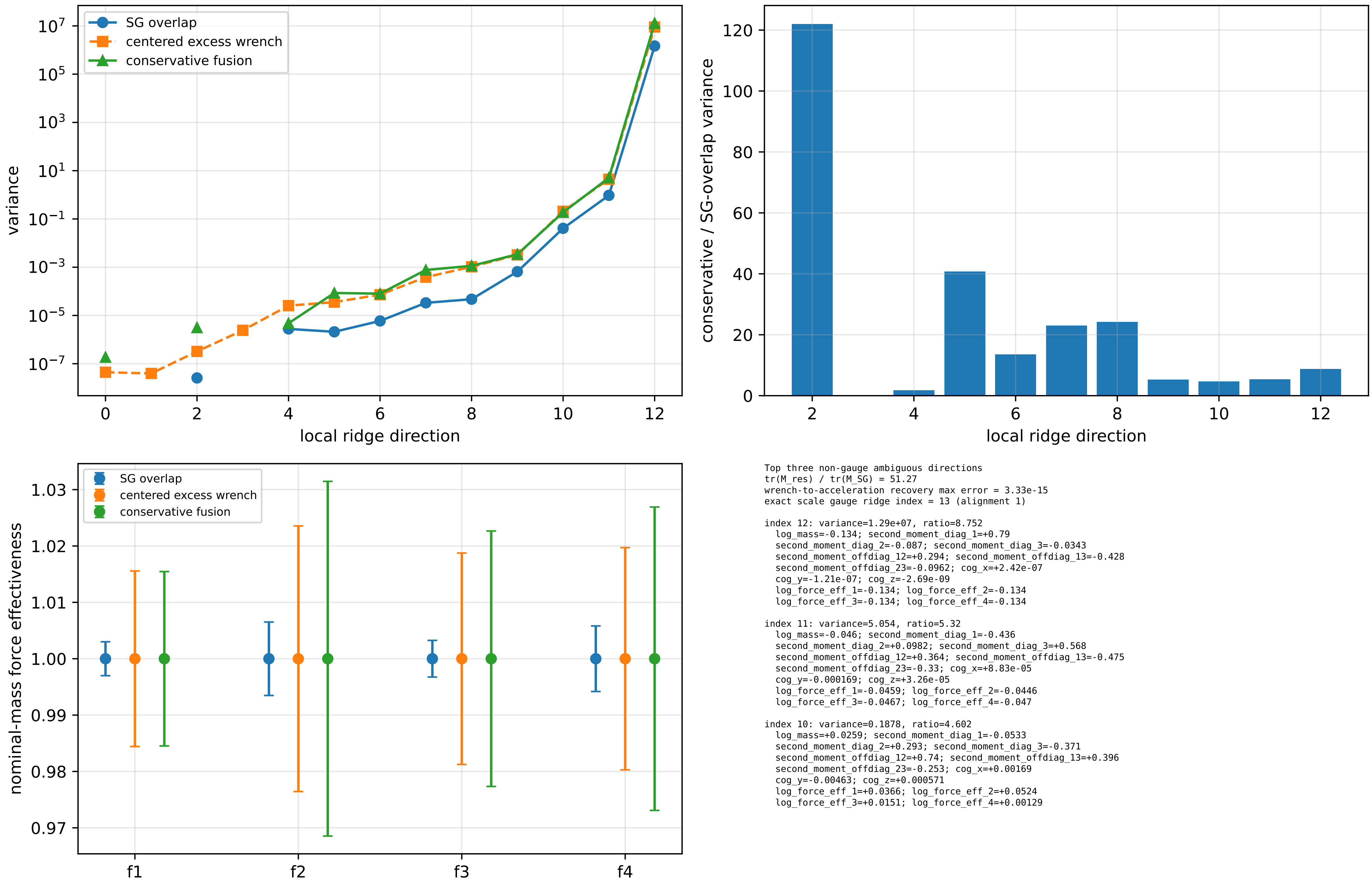
force_over_mass_all_nominal: parameter estimate

```
Case: force_over_mass_all_nominal
status: completed (all stages completed)

mass [kg]: nominal 2.35159759; estimated 2.8663266
inertia [kg m^2]:
[[ 3.55826878e-01  4.52130408e-05 -6.88693189e-02]
 [ 4.52130408e-05  2.80099323e-01 -1.99746384e-02]
 [-6.88693189e-02 -1.99746384e-02  5.97149628e-01]]
CoG body [m]: [-0.00762164  0.00962124 -0.04995782]
force effectiveness: [1.21888279 1.21888299 1.21888282 1.21888279]

scale-free J/m [m^2]:
[[ 1.24140382e-01  1.57738622e-05 -2.40270313e-02]
 [ 1.57738622e-05  9.77206586e-02 -6.96872382e-03]
 [-2.40270313e-02 -6.96872382e-03  2.08332724e-01]]
scale-free f/m: [0.42524212 0.42524219 0.42524213 0.42524212]

rotor lag cell: (0.19297576, 0.19300127] s
representative: 0.192988515 s
```

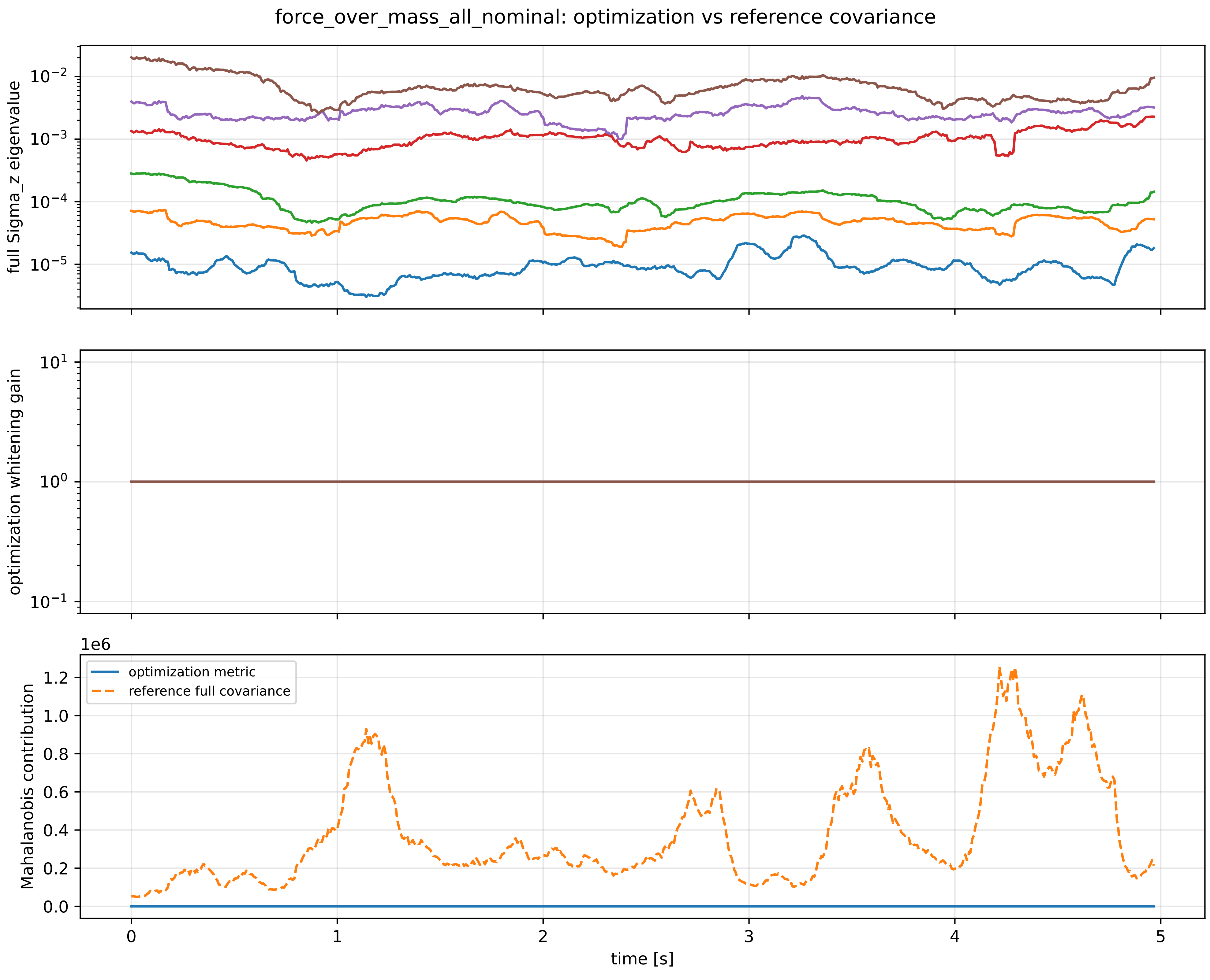
force_over_mass_all_nominal: ridge spectrum (rank 13, nullity 1, ||Jv||=2e-13)



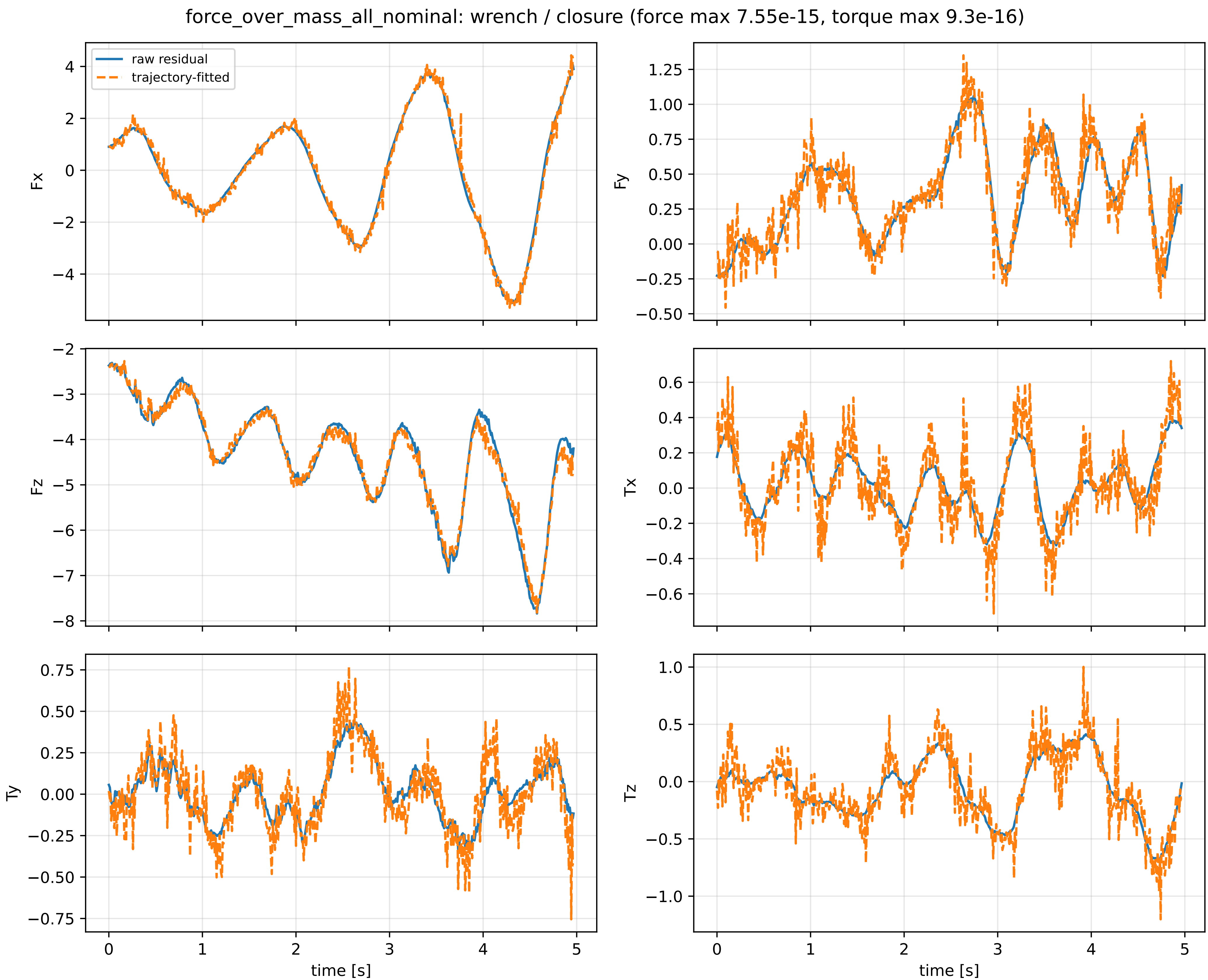
force_over_mass_all_nominal: Conservative fusion uncertainty



The conservative fusion covariance deliberately retains the existing SG-overlap uncertainty and adds the uncentered empirical residual score second moment without subtracting the SG contribution. It is intended as a conservative fusion distribution, not as a calibrated generative noise covariance.



force_over_mass_all_nominal: optimization vs reference covariance
full Sigma_z eigenvalue
optimization whitening gain
Mahalanobis contribution
1e6
optimization metric
reference full covariance
time [s]



force_over_mass_all_nominal: wrench / closure (force max 7.55e-15, torque max 9.3e-16)
raw residual
trajectory-fitted
Fx
Fy
Fz
Tx
Ty
Tz
time [s]
time [s]

force_over_mass_all_nominal: optional parameter-prior audit

```
Optional physical parameter prior

The parameter prior is optional external information. The default estimator is prior-free.
name: pseudo_condition_force_over_mass_all_nominal
role: pseudo_conditioning_ablation
source: /home/leus/catkin_ws/src/ProbTF-demo/ros/examples/grape-param-estim/minimal/config/priors/pseudo_conditioning/group/force_over_mass_
SHA256: 1299af3479d15bf1e5d1f52aba937e63fa8162cc154062ba161787f233f29dd2

data objective:  1138.72175487
prior objective: 0.00945680610117
total objective: 1138.73121167

This configuration uses an intentionally tight artificial standard deviation
for an ablation-style pseudo-conditioning experiment; it is not a calibrated
physical uncertainty.

factor: force_over_mass_all_nominal (force_effectiveness_over_mass)
  components: rotor_1, rotor_2, rotor_3, rotor_4
  target: [0.42524282 0.42524282 0.42524282 0.42524282]
  std:    [1.e-05 1.e-05 1.e-05 1.e-05]
  final:  [0.42524212 0.42524219 0.42524213 0.42524212]
  error:  [-7.06078075e-07 -6.38214288e-07 -6.97608467e-07 -7.06286005e-07]
  standardized residual: [-0.07060781 -0.06382143 -0.06976085 -0.0706286 ]
  factor objective: 0.00945680610117
```